SCHOOL OF COMPUTER SCIENCE ENGINEERING | MRCET

(CSE)

Email: mrcetcse2004@gmail.com

IV Year B.Tech-II Semester Major Project Individual Summery Sheet

Project Title:	**An Agentic AI-Based Secure Document Management System**				
Project Code:	R22A0595	**Batch Size:**	3	**Batch:**	2022-26
Domain / Area:	Artificial Intelligence & Blockchain Security Systems		**SDG Mapping**	9	

Abstract:

TrustAgent is an AI-based secure document management system designed to protect and control access to sensitive user documents. The system allows users to upload personal documents and share them using time-limited, revocable links. An intelligent agent interprets user commands and autonomously manages document sharing workflows. Blockchain technology ensures immutability and tamper-proof logging of document transactions, while IPFS enables secure decentralized storage. This project improves privacy, security, and trust in digital document sharing.

Technical (S/w & H/w) Specifications

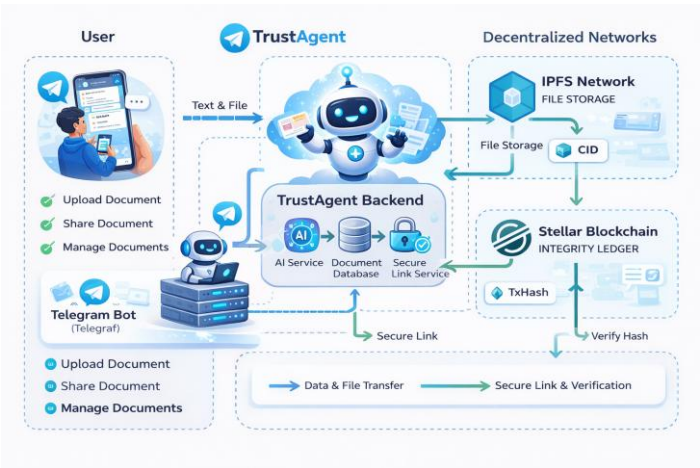
Software:
Frontend: React.js
Backend: Node.Js
Database: MongoDb
Blockchain: Stellar
Storage: IPFS

Hardware:
Processor: Inteli5
Storage: 50Gb

Module(s) Specifications

Module 1: User Interaction Module
Module 2: AI Intent & Document Classification
Module 3: Secure Link & Access Control Module
Module 4: Secure Document Storage Module (IPFS)
Module 5: Blockchain Verification Module

Architecture Diagram



Methodology

The TrustAgent system allows users to upload documents through a Telegram bot or web interface. The uploaded document is analyzed using AI to identify its type and intent. Documents are securely stored using decentralized IPFS storage, and a blockchain-based reference is created to ensure immutability and verification. Secure, time-limited access links are generated for document sharing, preventing unauthorized access. Admin controls and access logs enable monitoring, revocation, and auditing of document usage, ensuring privacy and trust.

Existing System

1. Insecure Document Sharing
2. No Access Control or Expiry
3. Lack of Document Verification
4. Centralized Storage Risks
5. No Audit Trail or Transparency

Proposed System

1. Secure Document Upload & Storage
2. Agent-Based Interaction
3. Time-Bound Access Control
4. Blockchain-Backed Integrity
5. Fine-Grained User Control

Guide Details

Dr D RADHA
Associate Professor
Dept. of CSE

MRCET Campus
UGC Autonomous Institution-
Govt. of India

Batch Members Details

Pattela Yuva Akhil
22N31A05G9

P. Bhargava Sai
22N31A05J9

Md. Irfan
22N31A05E0